

**THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NJOMBE REGION
FORM SIX PRE-MOCK EXAMINATION**

143/2

MATHEMATICS 2

(For Both School and Private Candidates)

Duration: 3 Hours

Monday 24th August 2026

Instructions

1. This paper consists of sections A and B with a total of **eight (08)** questions.
2. Answer **all** questions in section A and **two (2)** questions from section B.
3. All work done in answering each question must be shown clearly.
4. NECTA's Mathematical tables and non-programmable calculator may be used.
5. Communication devices and any unauthorized materials are **not** allowed in the examination room.
6. Write your examination number on every page of your answer booklet(s)

SECTION A (60 Marks)

Answer **all** questions in this section.

1. (a) A medical Centre encourages elderly people to have a flu vaccination each year. The vaccination reduces the likelihood of getting flu from 40% to 10%. If 45% of the elderly people visiting the Centre have the vaccination, find the probability that an elderly person chosen at random.

- (i) Gets flu
- (ii) Had the vaccination, given that they get flu.

- (b) If X is a discrete random variable with probability distribution

X	1	2	3	4
P	a	0.3	b	0.2

Given that $E(x) = 2.7$, find the value of a and b .

- (c) A probability density function is given by

$$Q(x) = \begin{cases} \frac{x}{6}, & \text{for } 0 \leq x < 3 \\ \frac{4-x}{2}, & \text{for } 3 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Sketch the graph of $Q(x)$

- (ii) Calculate the probability that x occurs in the interval $[1,2]$.

2. (a) Calculate the angles of a triangle ABC with vertices A $(2, -3, 5)$, B $(0, 1, 4)$, and C $(-2, 5, 2)$. Where A, B and C are vectors respectively.

- (b) If $|\underline{a}| = 2$, $|\underline{b}| = 3$, and $\theta = 60^\circ$ find $|\underline{a} + 2\underline{b}|$

- (c) Two forces of magnitude 5 and 3 units acting in the direction of the vectors $6\underline{i} + 2\underline{j} + 3\underline{k}$ and $3\underline{i} - 2\underline{j} + 6\underline{k}$, respectively, act on a particle which is displaced from point $(2, 1, -3)$ to the point $(4, -3, 1)$. Find the work done by the forces.

3. (a) The roots of the equation $x^3 + ax^2 + bx + 30 = 0$ are in the ratio 2:3:5. Find the values of a and b .

- (b) Express $\frac{10-17x+14x^2}{(2+x)(1-2x)^2}$ into partial fractions. Hence obtain the expansion up to and including the term in x^3 .

(c) (i) Given that $A = \begin{pmatrix} 2 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 & 2 \\ 3 & 2 & 2 \\ 1 & 0 & 1 \end{pmatrix}$, compute $|AB|$

(ii) Use the result obtained in (c) (i) to solve the following

Simultaneous equations;

$$\begin{cases} 2x - y - 2z = 6 \\ -x + y = 1 \\ -2x + y + 3z = -7 \end{cases}$$

4. (a) (i) Given that $z = x + iy$, find the cartesian equation if $\arg(z + i) = \frac{\pi}{4}$.

(ii) If $z = a + ib$ and $z^5 = 1$, show that $4a(b^4 - a^4) = 1$.

(b) Use de Moivre's theorem to prove that $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$.

(c) Solve the equation $z^5 - 3i + 3 = 0$, express the roots in polar form.

SECTION B (40 Marks)

Answer **two (02)** questions in this section.

5. (a) If $\sin x + \sin y = a$ and $\cos x + \cos y = b$, show that

$$\cos^2 \frac{1}{2}(x - y) = \frac{1}{4}(a^2 + b^2).$$

(b) Solve the equation $\sin(x - 150^\circ) \cos(x + 150^\circ) = 0.5$ for $0^\circ \leq x \leq 360^\circ$.

(c) Draw the graph of $f(\theta) = \cos \theta$ for $-2\pi \leq \theta \leq 2\pi$.

(d) In any triangle ABC, prove that $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$.

6. (a) Find the general solution of $\frac{dy}{dx} + y = \sin x$.

- (b) Determine the values of p and q , for which the differential equation is $(y + x^3)dx + (px + qy^3)dy = 0$ is exact, hence find its general solution.
- (c) Find the solution of the differential equation:

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10\sin x.$$
Where $y(0) = \frac{5}{2}$, and $y'(0) = \frac{19}{2}$.
- (d) If the population of a country doubles in 50 years. In how many years will the population become three times the original? assuming that the rate of increase of the population is proportional to the number of inhabitants.
7. (a) If $ax + by + c = 0$ touches the parabola $x^2 = 4ay$, find the equation connecting a , b and c .
- (b) Write the equation of the ellipse $18x^2 + 12y^2 - 144x + 48y + 120 = 0$ in a standard form.
- (c) Transform the polar equation $r = \frac{4}{3+5\sin\theta}$ in cartesian form.
- (d) Sketch the graph of $r = 5\sin 3\theta$.